# NutriVision — AI-Driven Meal Analysis System

## System Requirements & Deployment Specifications

> **Introduction to Requirements & Deployment:**
> This section defines the core system requirements for **NutriVision** and translates the architecture into practical Deployment Diagrams. It covers both functional and non-functional dimensions, operational constraints, and establishes the interaction paths between client components (Flutter App) and backend/AI services (Firebase Cloud & External AI Analytics Engine) to ensure high reliability, privacy, and horizontal scalability.

## 1. Functional Requirements (FR)

| ID | Requirement | Description | Priority |
|---|---|---|---|
| FR-01 | User Registration | Create a new account using Name, Email, and Password. | **High** |
| FR-02 | User Login | Authenticate user credentials stored in Firestore via Firebase Auth. | **High** |
| FR-03 | Health Goals Management | Input and edit personal dietary and health goals (HealthGoals). | **High** |
| FR-04 | Meal Image Upload | Capture or select a meal image from the device gallery. | **High** |
| FR-05 | AI Meal Analysis | Send image/description to AI service and retrieve nutritional analysis. | **High** |
| FR-06 | Nutritional Data Display | Display detailed breakdown of Calories and Macronutrients (Macros). | **High** |
| FR-07 | Healthy Alternatives | Suggest healthy food alternatives based on the meal evaluation. | **Medium** |
| FR-08 | Meal Log (MealLog) | Save and view historical meal entries with photos and notes. | **High** |
| FR-09 | Recipe Exploration | Browse recipes categorized by meal type and nutrition profile. | **Medium** |
| FR-10 | Reports & Analytics | Generate periodic health and nutrition intake summary reports. | **Medium** |

## 2. Non-Functional Requirements (NFR)

| ID | Requirement | Description |
|---|---|---|
| NFR-01 | Performance | Meal analysis response time must remain under 5 seconds. |
| NFR-02 | Security | End-to-end data encryption in transit (Dio Interceptors + HTTPS) & Firebase Auth. |
| NFR-03 | Scalability | Leverage Firebase Cloud Firestore for automatic horizontal scaling. |
| NFR-04 | Platform Support | Cross-platform compatibility for iOS and Android via Flutter. |
| NFR-05 | User Experience | Smooth UI interaction with clear loading progress indicators during AI processing. |

| ID | Requirement | Description |
|---|---|---|
| NFR-06 | Reliability | Comprehensive error handling & retry mechanisms managed by Dio Interceptors. |
| NFR-07 | Maintainability | Clean separation of architectural layers (UI / Business Logic / Data / AI Services). |
| NFR-08 | Privacy | User meal images strictly kept private and isolated from external analytics sharing. |

## 3. System Constraints

- Backend infrastructure strictly relies on Firebase Backend-as-a-Service (BaaS).
- AI Service operates via Prompt Engineering (Search & Discovery) rather than a locally hosted custom model.
- Application UI built in Flutter utilizing layout structures such as GridView and Stack.

## 4. High-Level Deployment Diagram

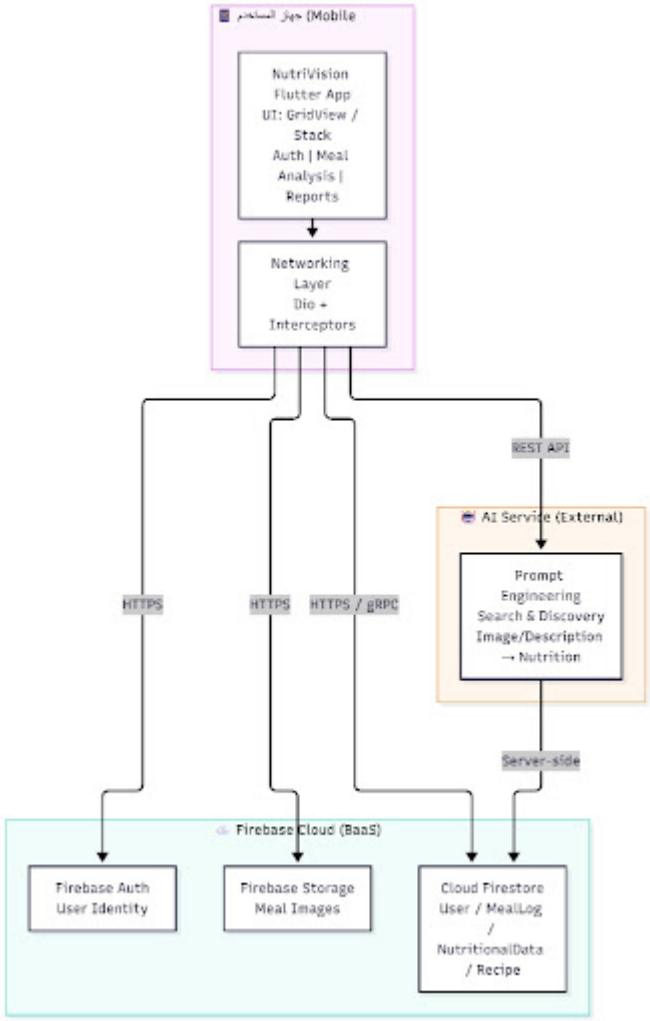



*Figure 1: High-Level Deployment Diagram depicting Client-to-Cloud relationships.*

## 5. Detailed Deployment Diagram (Nodes & Artifacts)

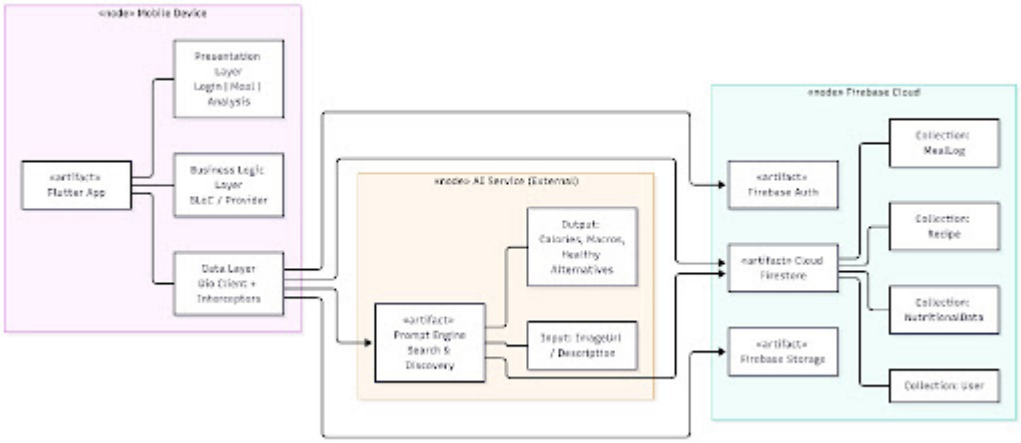



*Figure 2: Detailed Deployment Diagram showing system nodes, artifacts, and database collections.*

## 6. Communication Sequence & Network Paths

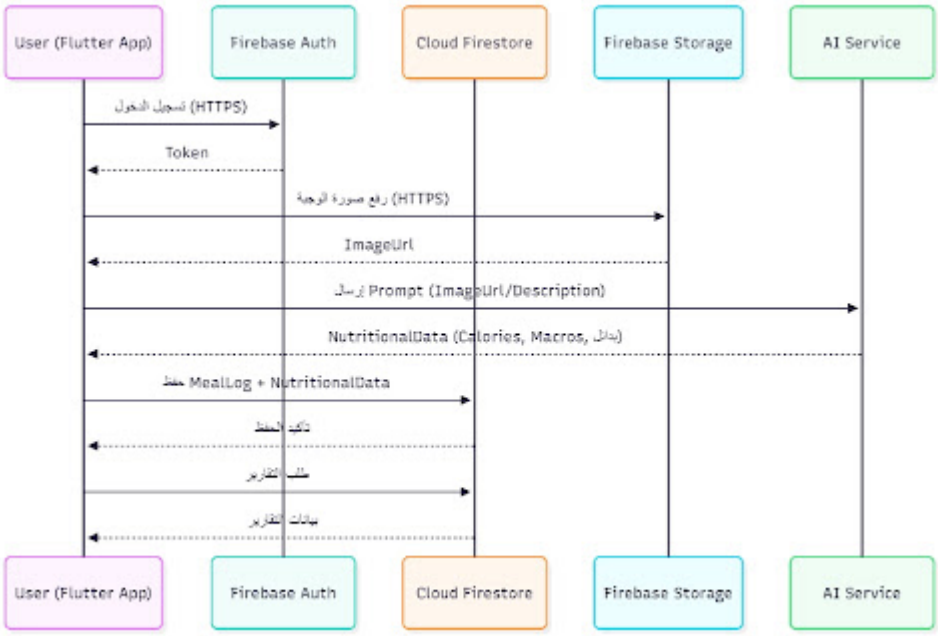



*Figure 3: Sequence Diagram illustrating message flow and communication protocols between components.*

### Communication Channels Overview:

| From | To | Protocol | Transferred Payload / Data |
|---|---|---|---|
| Flutter App | Firebase Auth | HTTPS | User Login & Registration Credentials |
| Flutter App | Cloud Firestore | HTTPS / gRPC | User, MealLog, NutritionalData, Recipe documents |
| Flutter App | Firebase Storage | HTTPS | Upload / Retrieve Meal Image files |
| Flutter App | AI Service | HTTPS (REST via Dio) | Prompt (Image/Description) → NutritionalData |

| From | To | Protocol | Transferred Payload / Data |
|---|---|---|---|
| AI Service | Cloud Firestore | Server-side API | Processed Meal Analytics & Recommendations |

## 7. Requirement Traceability Matrix

| Requirement | Associated Entity | Target Module / Diagram Node |
|---|---|---|
| FR-01 / FR-02 | User | Deployment — Firebase Auth Node |
| FR-03 | User.HealthGoals | Cloud Firestore — User Collection |
| FR-04 / FR-05 | MealLog.ImageUrl | Firebase Storage + External AI Service Node |
| FR-06 | NutritionalData | Cloud Firestore — NutritionalData Collection |
| FR-07 | NutritionalData (Alternatives) | External AI Analytics Engine |
| FR-08 | MealLog | Cloud Firestore — MealLog Collection |
| FR-09 | Recipe | Cloud Firestore — Recipe Collection |
| FR-10 | NutritionalData + MealLog | Client Presentation Layer (Reports Module) |

**Conclusion & Integration:**
This document serves as the bridging specification between client UI/UX and cloud/AI infrastructure. The deployment diagrams, sequence flows, and traceability matrix confirm system readiness for development and high-capacity production deployment.